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# Trainee doctors' views on workplace-based assessments: Are they just a tick box exercise?

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## Abstract

**Background:** In August 2007, run-through specialty training and workplace-based assessments (WBAs) were introduced for UK junior doctors.

**Aim:** This study explores whether WBAs have been successfully integrated into paediatric training programmes.

**Method:** A 20-item questionnaire was distributed to regional paediatric trainees.

**Results:** Eighty seven per cent (130/150) trainees responded. Reliability of the study was good (Cronbach's alpha 0.813); 92% (119/130) trainees knew how to conduct WBAs (scoring  $\geq 4$  on the six-point Likert scale). However, they were ambivalent about WBAs being a true reflection of their capabilities (mean Likert score 3.46). A common problem was finding assessors with 40% of trainees stating that staff had refused to do assessments. Almost half of the assessments were done retrospectively. Trainees did receive feedback but advice on future improvement was not always given (highest mean Likert score 4.00). Free text comments analysis mostly centred on difficulties in getting WBAs done.

**Conclusions:** A cultural change is needed for trainees to feel that WBAs are not just a tick box exercise, but a useful educational tool for learning. Ongoing work on implementation needs to include additional training, especially on the value of WBAs for formative assessment and consultants having protected time in their job plans for training.

## Introduction

In the UK, there have been significant changes to postgraduate training with the implementation of Modernising Medical Careers (MMC) and a competency-based assessment process for junior doctors (Department of Health 2003, 2004). The Postgraduate Medical Education and Training Board (PMETB), an independent professional body which regulated UK postgraduate medical education and training, drew up the principles and standards for the framework of the assessment process (Southgate & Grant 2003). From 1 April 2010, PMETB merged with the General Medical Council (GMC) and the GMC is now responsible for regulating all stages of UK medical education.

Currently, trainees undertake a 2-year foundation programme followed by specialty training. Progress is dependent on the achievement of competencies followed by competitive selection into speciality training. This is a 'run-through' (coupled or continuous) programme in Paediatrics whereas some specialties have further competitive interviews to progress from basic to higher specialty training (uncoupled programme). Traditionally, assessment of trainee doctors was mostly based on written examinations, i.e. what a person knows rather than what he or she does in practice. However, the attributes of doctors are multiple and include not only specialty knowledge and skills, but also generic skills and attitudes such as honesty, integrity, communication and team working. Competence can therefore only really be inferred from actual performance, i.e. the way the person performs and

## Practice points

- Consultants need allocated time in their job plans to conduct WBAs for trainees
- Trainees need to understand the role of WBAs
- Pre-planned opportunities for WBAs require collegial working between trainers and trainees
- Timely and quality feedback to trainees is essential including how to improve performance
- WBAs need to be used for formative assessment and as an aid to learning to avoid them being viewed by trainees as a 'tick box exercise'
- Regular review of how WBAs are conducted is a valuable exercise so as to highlight and address problems and ensure quality is maintained.

behaves in a day-to-day practice (Swanwick & Chana 2005). Such competencies are expected to be achieved in the clinical environment and are termed workplace-based assessments (WBAs). Guidance on implementing WBAs was produced jointly by PMETB and Academy of Medical Royal Colleges (AoMRC) in 2009. WBAs are carried out in both coupled and uncoupled specialty programmes.

The assessment of clinical competence using WBAs has been based on the work by George Miller who in the 1990s developed a framework for assessing clinical competence – Miller's pyramid. At the top of pyramid you have 'Action' which focuses on what occurs in practice rather than

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what happens in an artificial setting or test situation. WBAs target this highest level of the pyramid and collect information about how doctors perform in their normal practice (Dent & Harden 2009; Mohanna et al. 2010). Subsequent to the work by Miller (1990), Norcini and his colleagues developed the mini Clinical Evaluation Exercise (mini-CEX) which has been found to be a reliable tool, with reasonable feasibility and satisfaction (Norcini et al. 1995, 1997, 2003). The latter tool is one of four tools currently used in practice for WBAs:

- (1) Multi-source feedback (MSF): What healthcare colleagues observe about performance on a day-to-day basis.
- (2) Mini-CEX: Observation of what actually happens during encounters with patients. It can focus on history, diagnosis, management or explanation.
- (3) Direct observation of procedural skills (DOPS): Observation of trainee's ability to perform commonly required procedures and skills.
- (4) Case-based discussion (CbD): Explores why a trainee managed a particular case in a particular way.

All four of the WBAs listed above are currently used in UK paediatric training. Further details can be found on the Royal College of Paediatrics and Child Health (RCPCH) website (<http://www.rcpch.ac.uk/Training/Assessments-in-the-Workplace>). Paediatric assessments are marked on a seven-point scale ranging from 0 (unsafe) to 6 (well above expectations) and are marked in the presence of the trainee. They are used to assess acquisition of curriculum competencies and the yearly number of each of the WBAs varies according to year of training (RCPCH 2010b). An appropriately senior trainee or consultant can be the assessor for WBAs except for CbDs which should be assessed by a consultant. RCPCH guidance is to give immediate feedback following the assessment.

Using WBAs provide an opportunity to not only improve training and facilitate interaction with the trainee but also ultimately improve patient care (Davies 2005). It is important to identify trainees with difficulties as early as possible in training so that remedial training can be provided. Prior to MMC, such problems were frequently 'discovered' in the penultimate year of training or after appointment as a consultant (Wragg et al. 2003; PMETB & AoMRC 2009). In UK specialty training, both in coupled and uncoupled programmes, the WBAs are part of the portfolio evidence which is reviewed at a trainee's annual review of competence progression and a recommendation made as to whether the trainee has sufficient evidence to pass their training year, requires more evidence, training for specific competencies or has to repeat the year. However, assessing the performance of doctors whilst they are engaged in clinical practice is not without its problems. There are many challenges including time, training and finding an adequate number of assessors to make the assessment valid (Nair et al. 2008; Murphy et al. 2009; Weller et al. 2009).

The aims of this study were to explore whether WBAs: mini-CEXs, DOPS and CbDs had been successfully integrated into run-through paediatric speciality training programmes as they were new assessments of paediatric trainees. MSF was well established in our Deanery in the form of Team Assessment Behaviour (Whitehouse et al. 2007). The study

looked at (1) whether trainees were knowledgeable about WBAs, (2) how WBAs were conducted in practice and (3) if WBAs were useful in medical training.

## Method

This study was conducted over a 2-month period (December 2008 to January 2009) in the West Midlands Deanery (WMD), UK. This is a large Deanery having approximately 10% of the UK junior doctors. Participants for the study were paediatric trainees who were either on run-through specialist training (ST) programmes or fixed term specialty training appointments (FTSTAs) of 1-year duration. The 20-item questionnaire (Figure 1) was designed along the principles by Oppenheim (1992) and Cohen et al. (2000). There were seven Likert scoring questions, 12 tick box questions of which six collected demographic data and a final free text question asking for any other comments on WBAs. The questionnaire was distributed to all paediatric trainees in the WMD at regional teaching days and to non-attendees by post and email. A follow-up email was sent to chase non-responders 4 weeks later. Ethical approval was not required as this is a service evaluation study (<http://www.corec.org.uk/informed-consent-policy.php>). All questionnaires were collected by the postgraduate school administrator and were anonymous.

## Data analysis

Data from all questionnaires were entered into an Excel spreadsheet and then transferred to the statistics programme SPSS version 15 for further analyses. Statistical analyses included frequencies, and descriptive statistics including means, ranges and standard deviations. Differences between groups in the data were analysed using ANOVA. Statistical significance was accepted with a *p* value of less than 0.05. Reliability as internal consistency in the seven Likert questions data was calculated using Cronbach's alpha. Thematic analysis was carried out according to the principles described by Coffey and Atkinson (1996) where themes emerge from the data. Themes were initially coded free nodes (by DWW) in the computer programme NVivo and then discussed by the three authors. These were then coded into free nodes and grouped into five overall themes.

## Results

There was an 87% (130/150) response rate to the questionnaire. There was good internal consistency in the data looking at the seven Likert questions from the respondents as a group with a Cronbach's alpha of 0.813. A value of 0.7 is generally regarded as acceptable for research study and a value of 0.8 for high stakes assessment (Howitt & Cramer 2003).

### Demographic data

Of the respondents, 69% (90/130) were female and 4% (5/130) worked part time. The majority (85%, 111/130) of trainees were aged 26–35 years (Table 1). Just under half of the trainees (49%, 64/130) had graduated in the UK. Nearly three-quarters

1. Are you aware of how work place based assessments should be conducted?

|                   |   |   |   |   |   |                           |
|-------------------|---|---|---|---|---|---------------------------|
| <b>Don't know</b> |   |   |   |   |   | <b>Very Knowledgeable</b> |
| 1                 | 2 | 3 | 4 | 5 | 6 |                           |

2. Have you had any training in mini-CEX, DOPS, Cbd? *Please tick all that apply*

a) None ☐

b) Deanery course on work based assessment ☐

c) E-learning package (DVD, web based training) ☐

d) Royal College documents ☐

e) Training provided by your trust ☐

f) Other please state ☐

3. **Have you been assessed** using work based assessments?

|                                                   |     |                          |    |                          |
|---------------------------------------------------|-----|--------------------------|----|--------------------------|
| a) Mini-CEX (Mini-Clinical Evaluation Exercise)   | Yes | <input type="checkbox"/> | No | <input type="checkbox"/> |
| b) DOPS (Direct Observation of Procedural Skills) | Yes | <input type="checkbox"/> | No | <input type="checkbox"/> |
| c) Cbd (Case Based Discussion)                    | Yes | <input type="checkbox"/> | No | <input type="checkbox"/> |

4. How confident are you that workplace based assessments are a true reflection of your capabilities?

|                      |   |   |   |   |                       |
|----------------------|---|---|---|---|-----------------------|
| <b>Not confident</b> |   |   |   |   | <b>Very confident</b> |
| 1                    | 2 | 3 | 4 | 5 | 6                     |

5. How easy is it to find assessors to carry out your workplace based assessments?

|                       |   |   |   |   |                  |
|-----------------------|---|---|---|---|------------------|
| <b>Very difficult</b> |   |   |   |   | <b>Very easy</b> |
| 1                     | 2 | 3 | 4 | 5 | 6                |

6. Who does your workplace based assessments i.e. mini-CEX, DOPS, Cbd?

*Please circular for each of the following for frequency*

|                         |              |   |   |   |                        |
|-------------------------|--------------|---|---|---|------------------------|
|                         | <b>Never</b> |   |   |   | <b>Very frequently</b> |
| Consultant              | 1            | 2 | 3 | 4 | 5 6                    |
| SpR/ST4/ST5             | 1            | 2 | 3 | 4 | 5 6                    |
| Staff grade             | 1            | 2 | 3 | 4 | 5 6                    |
| Sister                  | 1            | 2 | 3 | 4 | 5 6                    |
| Nursing staff           | 1            | 2 | 3 | 4 | 5 6                    |
| Other please state..... | 1            | 2 | 3 | 4 | 5 6                    |

7. How are your workplace based assessments planned?

*Please tick one option for each of the following assessments*

|                    |                                            |                |                                            |               |                                            |
|--------------------|--------------------------------------------|----------------|--------------------------------------------|---------------|--------------------------------------------|
| a) <b>Mini-CEX</b> | Pre planned <input type="checkbox"/>       | b) <b>DOPS</b> | Pre planned <input type="checkbox"/>       | c) <b>Cbd</b> | Pre planned <input type="checkbox"/>       |
|                    | Ad hoc/on the job <input type="checkbox"/> |                | Ad hoc/on the job <input type="checkbox"/> |               | Ad hoc/on the job <input type="checkbox"/> |
|                    | Retrospective <input type="checkbox"/>     |                | Retrospective <input type="checkbox"/>     |               | Retrospective <input type="checkbox"/>     |
|                    | Other please state.....                    |                | Other please state.....                    |               | Other please state.....                    |

8. How soon after the assessment do you get feedback?

*Please tick one option for each of the following assessments*

|                    |                                     |                |                                     |               |                                     |
|--------------------|-------------------------------------|----------------|-------------------------------------|---------------|-------------------------------------|
| a) <b>Mini-CEX</b> | < 5 mins <input type="checkbox"/>   | b) <b>DOPS</b> | < 5 mins <input type="checkbox"/>   | c) <b>Cbd</b> | < 5 mins <input type="checkbox"/>   |
|                    | 6-10 mins <input type="checkbox"/>  |                | 6-10 mins <input type="checkbox"/>  |               | 6-10 mins <input type="checkbox"/>  |
|                    | 11-20 mins <input type="checkbox"/> |                | 11-20 mins <input type="checkbox"/> |               | 11-20 mins <input type="checkbox"/> |
|                    | 21-30 mins <input type="checkbox"/> |                | 21-30 mins <input type="checkbox"/> |               | 21-30 mins <input type="checkbox"/> |
|                    | 31-60 mins <input type="checkbox"/> |                | 31-60 mins <input type="checkbox"/> |               | 31-60 mins <input type="checkbox"/> |
|                    | 1-2 hrs <input type="checkbox"/>    |                | 1-2 hrs <input type="checkbox"/>    |               | 1-2 hrs <input type="checkbox"/>    |
|                    | > 2 hrs <input type="checkbox"/>    |                | > 2 hrs <input type="checkbox"/>    |               | > 2hrs <input type="checkbox"/>     |

9. How long does it take for the assessment to be performed including feedback?

*Please tick one option for each of the following assessments*

|                    |                                     |                |                                     |               |                                     |
|--------------------|-------------------------------------|----------------|-------------------------------------|---------------|-------------------------------------|
| a) <b>Mini-CEX</b> | < 5 mins <input type="checkbox"/>   | b) <b>DOPS</b> | < 5 mins <input type="checkbox"/>   | c) <b>Cbd</b> | < 5 mins <input type="checkbox"/>   |
|                    | 6-10 mins <input type="checkbox"/>  |                | 6-10 mins <input type="checkbox"/>  |               | 6-10 mins <input type="checkbox"/>  |
|                    | 11-15 mins <input type="checkbox"/> |                | 11-15 mins <input type="checkbox"/> |               | 11-15 mins <input type="checkbox"/> |
|                    | 16-20 mins <input type="checkbox"/> |                | 16-20 mins <input type="checkbox"/> |               | 16-20 mins <input type="checkbox"/> |
|                    | 21-25 mins <input type="checkbox"/> |                | 21-25 mins <input type="checkbox"/> |               | 21-25 mins <input type="checkbox"/> |
|                    | 26-30 mins <input type="checkbox"/> |                | 26-30 mins <input type="checkbox"/> |               | 26-30 mins <input type="checkbox"/> |
|                    | > 30 mins <input type="checkbox"/>  |                | > 30 mins <input type="checkbox"/>  |               | > 30 mins <input type="checkbox"/>  |

10. How soon after the assessment does the paper work/online assessment form get completed?

*Please tick one option for each of the following assessments*

|                    |                                         |                |                                         |               |                                         |
|--------------------|-----------------------------------------|----------------|-----------------------------------------|---------------|-----------------------------------------|
| a) <b>Mini-CEX</b> | Immediately <input type="checkbox"/>    | b) <b>DOPS</b> | Immediately <input type="checkbox"/>    | c) <b>Cbd</b> | Immediately <input type="checkbox"/>    |
|                    | Within 2 hrs <input type="checkbox"/>   |                | Within 2 hrs <input type="checkbox"/>   |               | Within 2 hrs <input type="checkbox"/>   |
|                    | Same day <input type="checkbox"/>       |                | Same day <input type="checkbox"/>       |               | Same day <input type="checkbox"/>       |
|                    | Within 1 week <input type="checkbox"/>  |                | Within 1 week <input type="checkbox"/>  |               | Within 1 week <input type="checkbox"/>  |
|                    | Within 1 month <input type="checkbox"/> |                | Within 1 month <input type="checkbox"/> |               | Within 1 month <input type="checkbox"/> |
|                    | Over 1 month <input type="checkbox"/>   |                | Over 1 month <input type="checkbox"/>   |               | Over 1 month <input type="checkbox"/>   |

**Figure 1.** Study questionnaire.

11. How often is the box for a plan for future improvement completed?  
Please ring one option for each of the following assessments
- |          | Never |   |   |   |   | Always |
|----------|-------|---|---|---|---|--------|
| Mini-CEX | 1     | 2 | 3 | 4 | 5 | 6      |
| DOPS     | 1     | 2 | 3 | 4 | 5 | 6      |
| CBD      | 1     | 2 | 3 | 4 | 5 | 6      |
12. Have you had a member of medical or nursing staff refuse to carry out an assessment? Yes ☐ No ☐  
If yes what are the reasons? Please tick all that apply
- a) Too busy ☐
- b) Not confident ☐
- c) Not had training ☐
- d) Did not understand the paperwork/online form ☐
- e) Other please state .....
13. How useful are workplace based assessments for helping with your medical training?  
Please ring one option for each of the following assessments
- |          | Not useful |   |   |   |   | Very useful |
|----------|------------|---|---|---|---|-------------|
| Mini-CEX | 1          | 2 | 3 | 4 | 5 | 6           |
| DOPS     | 1          | 2 | 3 | 4 | 5 | 6           |
| CbD      | 1          | 2 | 3 | 4 | 5 | 6           |
14. Does your educational supervisor review your workplace based assessments at appraisal meetings?
- |  | Never |   |   |   |   | Always |
|--|-------|---|---|---|---|--------|
|  | 1     | 2 | 3 | 4 | 5 | 6      |
- Now a few questions about yourself**
15. Are you Male ☐ Female ☐
16. Are you UK graduate ☐ EU graduate ☐ Non-EU graduate ☐
17. Your age 23-25 years ☐ 26-30 years ☐ 31-35 years ☐ 36-40 years ☐ > 41 years ☐
18. Do you work Full time ☐ Part time ☐
19. Which grade of training in paediatrics are you in?
- ST1 ☐ ST2 ☐ ST3 ☐ ST4 ☐ ST5 ☐
- FTSTA 1 ☐ FTSTA 2 ☐ FTSTA 3 ☐
20. Please write down any comments about workplace based assessments

Figure 1. Continued.

of trainees (72%, 94/130) were in level one training ST years 1–3 and FTSTA.

### Knowledge of WBAs

About 92% (119/130) of the trainees knew how WBAs should be conducted (scoring 4 or more on the 6-point Likert scale); 78% trainees reported that they had training in the use of WBAs (Table 2). Most had received training from the Royal College of Paediatric and Child Health (RCPCH) documents (38%, 49/130) and e-learning package (31%, 40/130). Small numbers of trainees had attended training provided either by the hospital trust (9%, 11/130) or attending a course organised by the WMD (5%, 6/130). Other forms of training (14%, 18/130) included word of mouth and training from peer colleagues. Worryingly, 22% (29/130) of trainees reported that they had received no training. All respondents had been assessed using all three types of WBAs.

### How WBAs are conducted

Respondents found difficulty in finding assessors to do the assessments (mean Likert score 2.38) (Table 2).

**Table 1.** Demographic data for trainees who completed the questionnaire on WBAs.

|                               | No of trainees (%)<br>Total = 130 |
|-------------------------------|-----------------------------------|
| <b>Age</b>                    |                                   |
| 23–25 years                   | 9 (7)                             |
| 26–30 years                   | 82 (63)                           |
| 31–35 years                   | 29 (22)                           |
| 36–40 years                   | 9 (7)                             |
| Not stated                    | 1 (1)                             |
| <b>Place of qualification</b> |                                   |
| UK                            | 64 (49)                           |
| Outside the EU                | 58 (45)                           |
| EU graduates                  | 8 (6)                             |
| <b>Year of training</b>       |                                   |
| ST1                           | 37 (28.5)                         |
| ST2                           | 27 (21)                           |
| ST3                           | 17 (13)                           |
| ST4                           | 24 (18.5)                         |
| ST5                           | 12 (9)                            |
| FTSTA1-3                      | 13 (10)                           |

Notes: EU, European union; ST, specialty training; FTSTA, fixed term specialty training appointment. Year of training is denoted by the number after ST or FTSTA.

**Table 2.** Responses to Likert questions on WBAs.

| Question                                                                                            | Mean Likert score                              | 95% confidence interval for mean |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------|
| Q1. Aware of how WBAs should be conducted                                                           | 4.52 (1 = do not know; 6 = very knowledgeable) | 4.39–4.65                        |
| Q4. Confidence that WBAs are a true reflection of your capabilities                                 | 3.46 (1 = not confident; 6 = very confident)   | 3.27–3.65                        |
| Q5. Ease of finding assessors to conduct WBAs                                                       | 2.38 (1 = very difficult; 6 = very easy)       | 2.20–2.55                        |
| Q6. Staff who conduct WBAs                                                                          |                                                |                                  |
| Consultant                                                                                          | 3.33                                           | 3.11–3.54                        |
| Senior trainee – SpR/ST4/ST5                                                                        | 4.21                                           | 3.97–4.45                        |
| Staff grade                                                                                         | 2.67                                           | 2.38–2.96                        |
| Sister                                                                                              | 1.96                                           | 1.76–2.17                        |
| Nursing staff                                                                                       | 1.90 (1 = never; 6 = very frequently)          | 1.68–2.11                        |
| Senior trainees were the most frequent assessors ( $p < 0.001$ )                                    |                                                |                                  |
| Q11. Box for future improvements completed                                                          |                                                | 3.52–3.98                        |
| Mini-CEX                                                                                            | 3.75                                           | 3.24–3.72                        |
| DOPS                                                                                                | 3.48                                           | 3.77–4.22                        |
| CbD                                                                                                 | 4.00 (1 = never; 6 = always)                   |                                  |
| The box for future improvement was completed significantly more frequently for CbDs ( $p < 0.001$ ) |                                                |                                  |
| Q13. Usefulness of WBAs in medical training                                                         |                                                |                                  |
| Mini-CEX                                                                                            | 3.23                                           | 3.02–3.45                        |
| DOPS                                                                                                | 3.20                                           | 2.96–3.44                        |
| CbD                                                                                                 | 3.70 (1 = not useful; 6 = very useful)         | 3.47–3.94                        |
| CbDs were the most useful WBA ( $p < 0.001$ )                                                       |                                                |                                  |
| Q14. Educational supervisor reviews WBAs at appraisal meetings                                      | 4.10 (1 = never; 6 = always)                   | 3.87–4.34                        |

Higher specialist trainees (specialist registrar (SpR)/ST4 or 5) were more likely to undertake the WBAs for their more junior colleagues (mean Likert score 4.21) in contrast to nursing staff (mean Likert score 1.90). It was reassuring that ST5 trainees' assessments were being undertaken by consultants which significantly differed from the assessor for those in ST1 who was a more senior trainee ( $p < 0.001$ ).

About 40% (52/130) of the respondents reported that a member of medical or nursing staff had refused to do an assessment. Reasons in order of frequency were being too busy (85%, 44/52), did not understand paperwork or online form (38%, 20/52), lack of training (17%, 9/52) and not being confident (12%, 6/52). Other reasons (27%, 14/52) included staff refusing to do the assessments because they did not like doing them or not having the password to do the assessment online.

#### How WBAs were conducted

Preplanning of assessments was best for CbDs, but was still only done in around a third (32%, 42/130). In around 80% of WBAs, the time taken to perform them was  $\leq 20$  min. There was a delay in feedback being given after WBAs by  $>1$  h for around 15% of trainees. In around a third of the WBAs, assessment forms either online or paperwork were not completed within a week and in some cases (15%) the time taken was over 1 month (Table 3).

#### Usefulness of WBAs in medical training

A mean Likert score of 3.46 was given by respondents about how confident they felt that WBAs were a true reflection of

their capabilities as a doctor. The usefulness of the different types of WBAs can be seen in Table 2 along with how often the box for future improvements was completed by assessors. Review of WBAs by educational supervisors at appraisal meetings was done with a mean score of 4.10.

Perception of WBAs varied by place of graduation and age. UK graduates found mini-CEXs and DOPS less useful compared to overseas graduates ( $p < 0.001$ ). Trainees who were older found mini-CEXs and DOPS more useful ( $p < 0.001$ ) compared to younger aged trainees. Those who had training in WBAs were more likely to have the box for improvements completed for the three types of assessments; mini-CEX ( $p = 0.022$ ), DOPS ( $p = 0.022$ ) and CbD ( $p = 0.001$ ). There was no significant difference found for gender, working full or part time, or year of training.

#### Free text comments

Sixty per cent (78/130) of trainees gave free text comments on the use of WBAs. Analysis revealed 15 themes which was grouped into five categories, the most common being difficulties in conducting assessments, as well as performance and usefulness of assessments (Table 4).

## Discussion

Assessing the performance of trainee doctors in the workplace clinical environment is important yet challenging for both trainees and trainers. WBAs provide a snap shot of a single clinical encounter. Optimum assessment is by different assessors directly observing the trainee engaging with patients who have a variety of clinical problems and in different



**Table 3.** How WBAs are conducted.

|                                                                                                                                                 | Mini-CEX n/130 (%) | DOPS n/130 (%) | CbD n/130 (%) |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|---------------|
| Planning of WBAs                                                                                                                                |                    |                |               |
| Pre Planned                                                                                                                                     | 23 (18)            | 6 (5)          | 42 (32)       |
| Ad hoc/on the job                                                                                                                               | 69 (53)            | 82 (63)        | 29 (22)       |
| Retrospective                                                                                                                                   | 38 (29)            | 42 (32)        | 59 (46)       |
| There was no significant difference in planning between the 3 WBAs ( $p = 0.067$ )                                                              |                    |                |               |
| Time taken after WBA to receive feedback                                                                                                        |                    |                |               |
| ≤10 min                                                                                                                                         | 92 (71)            | 98 (75)        | 92 (71)       |
| 11–20 min                                                                                                                                       | 13 (10)            | 10 (8)         | 11 (8)        |
| 21–30 min                                                                                                                                       | 7 (5)              | 4 (3)          | 10 (8)        |
| 31–60 min                                                                                                                                       | 0 (0)              | 2 (2)          | 1 (1)         |
| ≥1 h                                                                                                                                            | 18 (14)            | 16 (12)        | 16 (12)       |
| DOPS feedback was significantly quicker than for mini-CEXs and CbDs ( $p < 0.001$ )                                                             |                    |                |               |
| Time to complete assessment including feedback                                                                                                  |                    |                |               |
| ≤10 min                                                                                                                                         | 36 (28)            | 54 (42)        | 16 (12)       |
| 11–20 min                                                                                                                                       | 70 (54)            | 64 (49)        | 59 (45)       |
| 21–30 min                                                                                                                                       | 17 (13)            | 9 (7)          | 32 (25)       |
| 30 min                                                                                                                                          | 7 (5)              | 3 (2)          | 23 (18)       |
| DOPS assessments were significantly quicker than for mini-CEXs and CbDs ( $p < 0.001$ )                                                         |                    |                |               |
| Completion of paperwork/online form following assessment                                                                                        |                    |                |               |
| ≤2 h                                                                                                                                            | 37 (29)            | 34 (26)        | 43 (33)       |
| Same day                                                                                                                                        | 18 (14)            | 19 (15)        | 20 (15)       |
| Within 1 week                                                                                                                                   | 39 (30)            | 36 (27)        | 23 (18)       |
| Within 1 month                                                                                                                                  | 16 (12)            | 22 (17)        | 22 (17)       |
| Over 1 month                                                                                                                                    | 20 (15)            | 19 (15)        | 22 (17)       |
| There was no significant difference between the three WBAs for time of completion of paperwork/online form following assessment ( $p = 0.871$ ) |                    |                |               |

**Table 4.** Thematic analysis of free text comments with examples.**Difficulties in getting conducting assessments** (121 comments)

- i. Online access difficult
- ii. Difficulty getting assessors
- iii. Lack of time to do assessments
- iv. Paper systems easier

*'There is always a problem getting assessments completed ONLINE as the assessors either struggles to remember their password or there is a technical problems'*

*'Consultants are very busy to allocate time for this. This becomes more difficult if they have to observe procedures (which can be a long time)'*

*'DOPS are extremely difficult to get as most procedures are emergencies and usually out of hours so there is no 'senior' consultant around to DOPS you. You can't wait for a consultant to come in just to be supervised'*

*'Paper forms are easier to fill in'*

**1. Performance of assessments** (59 comments)

- i. Reliability
- ii. Need for assessor training
- iii. Feedback
- iv. Done retrospectively

*'Massive variability of assessment scores depending on assessor'*

*'Doctors need more training in how to fill out the forms'*

*'There is no interest in using the opportunity for education and feedback'*

*'Usually end up being completed retrospectively due to time constraints and pressures of service provision'*

**2. Usefulness of assessments** (48 comments, useful 22, useless 26)

*'WBA are excellent tools to promote and monitor a trainee's progress'*

*'I feel that the aim of the assessments is good, in particular the CbDs when done properly are useful and DOPS are evidence of competence'*

*'Generally they are a waste of time for a good doctor. They won't be useful to differentiate good from bad doctors. Bad doctors can actually look better if they do these assessments infrequently'*

*'Soul destroying, lots of very patronising paperwork. How can this raise the standard of medicine? It lowers the goal posts and makes juniors less willing to try at procedures they aren't comfortable with'*

**3. Box ticking** (7 comments)

*'Now it is just so people can tick the boxes. The inspiration and pride in our profession has gone'*

**4. Stressful** (4 comments)

*'It makes me very stressed out especially as I am going to finish the post soon. It is not an exaggeration if to say I wake up every morning wishing that the assessment system gets scrapped off'*

clinical settings. The purpose of the performance assessment is to identify underperforming doctors early in training who need extra support provided through a supportive environment (PMETB & AoRMC 2009). It is therefore important that feedback on WBAs is obtained from trainees to ensure problems with assessments are highlighted and solutions found. This study is the first of its kind which seeks trainees' views on all three WBAs.

The trainees who participated in this study are representative of those who were on the run-through paediatric ST programme in the West Midlands. The majority of trainees were in year 1 and 2 of the run-through training programme with small numbers at ST5 which was the highest training grade. This is because the new form of postgraduate specialty training was introduced in August 2007 as part of MMC with the highest grade recruited to being ST4 and a number of higher speciality training posts were still occupied by the old SpR grade.

It was surprising to see that overseas graduates found mini-CEXs and DOPS more useful than the UK graduates, a finding which needs further exploration in future studies. It may be that overseas graduates find the WBAs a useful aid to learning, as they have previously worked in clinical environments outside the UK where emphasis has been on service delivery rather than competency-based training. The UK graduates, on the other hand, are expected to do WBAs in the foundation years, i.e. from the onset of postgraduate training. Prior experience of WBAs, either positive or negative, may influence a trainee's perception of undertaking WBAs in subsequent training. The UK graduates may see WBAs as a *tick box exercise* to show that competencies have been achieved in foundation training before applying for the UK speciality training.

Trainees found difficulty in getting WBAs completed in agreement with Carr (2005). This is perhaps not surprising as they do take a significant amount of time to perform if done properly with the majority in our study being completed within 20 min. Preplanning of WBAs was reported by less than a third of trainees for CbDs and even less for the other assessments. Planning and scheduling of DOPS and mini-CEXs into day-to-day clinical practice such as clinics and ward rounds is important so as to minimise the impact of WBA tools on clinical service. Nair et al. (2008) found that problems with rostering or patients being away from wards caused difficulties for WBAs being performed. Time pressures and difficulty in finding a consultant were highlighted in the free text comments, and this has been noted in previous studies (Nair et al. 2008; Cohen et al. 2009). It is vital that both trainees and assessors work together in negotiating a time which can be used effectively when conducting such assessments.

The range of competencies trainees need to acquire differs by level of training and can be found in the RCPCH curriculum (RCPCH 2010b). It is reassuring that senior trainees were assessed by consultants and not surprising that junior trainees tended to have more senior colleagues as assessors as they are more freely available than consultants. Length of time taken to perform WBAs noted in the free text comments is a recurring theme as to why it can be difficult to get a consultant to do the WBAs (Wilkinson et al. 2008). It is interesting to note that length of time for assessment and feedback was significantly

quicker for DOPS, and this may be due to the fact that each procedural skill is clearly defined whereas the other WBAs are more open ended.

It was pleasing to see that the majority of trainees received feedback within 20 min following assessment. Worryingly, up to 15% of trainees received feedback over 1 h after the assessment which is contrary to well-established principles that effective feedback needs to be timely (Cohen et al. 2009; Weller et al. 2009). Suggestions for future improvement of trainee performance were not always completed (mean Likert score 3.48–4.00). Such suggestions need to be incorporated into verbal feedback as well as written down as it is important that both good performances is acknowledged, and problems in clinical performance do not remain uncorrected for long periods of time (Norcini et al. 1995; Carr 2006). To enhance learning, both quality and timely feedback is required (Nair et al. 2008).

Over 80% of mini-CEXs and DOPS were undertaken and feedback given in 20 min or less which is speedier than reported in other studies with mean observation time for the mini-CEX being between 20 and 30.9 min with additional time for feedback ranging from 8 to 15.4 min (Hauer 2000; Kogan et al. 2003; Nair et al. 2008; Wilkinson et al. 2008). For CbDs, 87% trainees reported that feedback was done within 30 min after the assessment. This form of assessment often involves more discussion with the trainee and therefore, it is not surprising that they took longer.

Trainees reported that the most common reason why staff refused to do assessments was that of being too busy. Previous studies have raised concerns about the amount of assessor's time taken up by assessments (Wiles et al. 2007). In addition, senior trainees were more likely to conduct assessments than consultants. These points will however be addressed by implementation of the recommendations of the Professor Sir John Temple report into the impact of the European Working time directive on the quality of training (Temple 2010). Recommendations include consultants being more directly responsible for the delivery of 24/7 care, the need for consultant job plans to allow more flexible working to deliver high quality training, service delivery supporting training and recognising, developing and rewarding trainers.

Trainees did not feel confident that WBAs gave a true reflection of their capabilities

Possible reasons include lack of effective feedback following the assessment as mentioned above. An action plan as a means of facilitating self-directed learning for the trainee to improve performance should be included in feedback and this study showed that this did not always occur and was found previously by Holmboe et al. (2004) and Jackson and Wall (2010) in the use of the mini-CEX. Another possible explanation for lack of confidence in WBAs is the need to train assessors for scores to be accurate (Forrest et al. 2006; Cohen et al. 2009; Weller et al. 2009). Providing training will not only help assessors to understand the value of WBAs for trainees but ensure that they are done correctly. The GMC states that all assessors should undertake appropriate training and only assess in areas where they have competence (GMC



2010). In order to comply with the GMC standards, a 'training the trainers' course is compulsory in the WMD and the School of Paediatrics has run seven courses on assessment and appraisal in the last year attended by both consultants and senior trainees. However, this study highlights the need for compulsory training of other health professionals who may be approached by trainees as potential assessors.

Free text comments included some positive ones related to the usefulness of WBAs as a way of monitoring a trainee's progress which has been found in other studies (Wiles et al. 2007; Cohen et al. 2009). Of the three types of assessment, CbDs were particularly found to be useful for formative assessment due to one-to-one discussion about a case, a finding noted by Forrest et al. (2006). However, none of our trainees mentioned the educational benefit of WBAs.

It is of concern to have comments by 9% of trainees that WBAs were just a *tick box exercise* that needs to be completed in order to progress from 1 year to the next during training. This finding has been found by Jackson and Wall (2010) with the mini-CEX tool particularly where there was poor assessor engagement. Part of the problem, we feel is that there needs to be a cultural change about the way these assessments are being used in a summative way rather than as a means of formative assessment. Perhaps, workplace learning exercise may be a more appropriate term than assessment. In Paediatrics in the WMD, we have included training on the WBAs for all trainees. As trainees become more accustomed to doing WBAs, it should remove the stress and anxiety reported by some of our trainees which has also been noted in other studies (Wiles et al. 2007; Cohen et al. 2009).

Limitations to this study include the fact that this was the experience of paediatric trainees in one Deanery. However, the WMD has 10% of the UK trainees and a strong reputation for medical education. We believe that problems highlighted in this study are not unique to paediatrics and are likely to apply to any speciality where there are constraints of working in a busy stressful environment within a 48-working hour week. Further studies could compare the viewpoints of senior trainees who have completed the run-through training programme to those who have just started the training programme. Finally, this study focused on the trainee's perspective of using WBAs. Surveying the views of staff who are involved in the assessing is important, especially, if we are to encourage staff to conduct assessments within their day-to-day duties.

This study has provided invaluable feedback from trainees on WBAs, in particular that implementation has been challenging for trainees. Providing quality feedback in a timely manner and using the tools for formative assessment will enhance learning rather than just using them to provide evidence of satisfactory achievement and progress against GMC approved training curricula. However, in order to ensure quality assurance of the assessment process, it is vital that both resources and time are provided for implementation particularly by the Consultant workforce.

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